Homework 6 Solutions

Based on Lecture 6: Trigonometric Functions and Their Properties.

Angles without a degree symbol are in radians, and $k \in \mathbb{Z}$ means that k is any integer.

Section 1: A 120° Isosceles Triangle

Exercise 1. An isosceles triangle has equal sides of length 5 and vertex angle $120^\circ = 2\pi/3$. Find the base length and altitude from the vertex. Check the base length using the sine law.

Solution. The base angles are $(180^\circ - 120^\circ)/2 = 30^\circ$. The altitude from the vertex bisects the vertex angle and the base, so it creates a 30°-60°-90° triangle with hypotenuse 5 and a 60° angle at the vertex. Then

$$\text{altitude} = 5\cos 60^\circ = \boxed{\frac{5}{2}}, \qquad \frac{\text{base}}{2} = 5\sin 60^\circ = \frac{5\sqrt{3}}{2}, \qquad \boxed{\text{base} = 5\sqrt{3}}.$$

Check by the sine law, pairing the base a with $A = 120^\circ$ and a side of length 5 with the opposite 30° angle:

$$\frac{a}{\sin 120^\circ} = \frac{5}{\sin 30^\circ} = \frac{5}{1/2} = 10, \qquad a = 10 \cdot \frac{\sqrt{3}}{2} = 5\sqrt{3}.$$

This agrees. (Also $(5/2)^2 + (5\sqrt{3}/2)^2 = 25/4 + 75/4 = 25$.)

Exercise 2. Two sides of a triangle have lengths 3 and 5, and their included angle is $120^\circ = 2\pi/3$. Use the cosine law to find the third side exactly.

Solution. With $\cos(2\pi/3) = -1/2$,

$$c^2 = 3^2 + 5^2 - 2 \cdot 3 \cdot 5\cos\frac{2\pi}{3} = 9 + 25 - 30\left(-\frac{1}{2}\right) = 34 + 15 = 49.$$

Therefore $\boxed{c = 7}$.

Exercise 3. In triangle ABC, $A = 120^\circ = 2\pi/3$, $B = 45^\circ = \pi/4$, and the opposite side $a = 4\sqrt{3}$. Find angle C in degrees and radians, then use the sine law to find side b exactly.

Solution. The angles sum to 180°, so

$$C = 180^\circ - 120^\circ - 45^\circ = \boxed{15^\circ = \frac{\pi}{12}}.$$

By the sine law, $\dfrac{b}{\sin 45^\circ} = \dfrac{a}{\sin 120^\circ}$, so

$$b = \frac{4\sqrt{3} \cdot \sin 45^\circ}{\sin 120^\circ} = \frac{4\sqrt{3} \cdot \frac{\sqrt{2}}{2}}{\frac{\sqrt{3}}{2}} = \boxed{4\sqrt{2}}.$$

Check: $a/\sin A = 4\sqrt{3}/(\sqrt{3}/2) = 8$ and $b/\sin B = 4\sqrt{2}/(\sqrt{2}/2) = 8$.

Exercise 4. An isosceles triangle has side lengths 3, 3, and $3\sqrt{3}$. Use the cosine law to find the cosine of its vertex angle, then identify all three angles in degrees and radians.

Solution. The vertex angle A is between the equal sides and opposite the side $3\sqrt{3}$. The cosine law gives

$$(3\sqrt{3})^2 = 3^2 + 3^2 - 2 \cdot 3 \cdot 3\cos A, \qquad 27 = 18 - 18\cos A,$$

so $\cos A = (18 - 27)/18 = \boxed{-\tfrac{1}{2}}$. Since $0^\circ < A < 180^\circ$, we get $A = 120^\circ = 2\pi/3$. The base angles are equal, each $(180^\circ - 120^\circ)/2 = 30^\circ = \pi/6$. Thus the angles are

$$\boxed{120^\circ = \tfrac{2\pi}{3}, \quad 30^\circ = \tfrac{\pi}{6}, \quad 30^\circ = \tfrac{\pi}{6}}.$$

Check with Exercise 1's scaling: the sides $2, 2, 2\sqrt{3}$ of the lecture triangle are multiplied by $3/2$.

Section 2: Calculating $\sin 72^\circ$ Exactly

Exercise 5. Let $d = (1+\sqrt{5})/2$. Verify that $d^2 = d+1$ and $1/d = d-1$. Use the second identity to rewrite $\cos(2\pi/5) = 1/(2d)$ as $(\sqrt{5}-1)/4$.

Solution. First,

$$d^2 = \frac{(1+\sqrt{5})^2}{4} = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2}, \qquad d+1 = \frac{1+\sqrt{5}}{2}+1 = \frac{3+\sqrt{5}}{2},$$

so $\boxed{d^2 = d+1}$. Dividing $d^2 = d+1$ by d gives $d = 1 + 1/d$, hence $\boxed{1/d = d-1}$. Directly, $d - 1 = \frac{1+\sqrt{5}-2}{2} = \frac{\sqrt{5}-1}{2}$ and

$$\frac{1}{d} = \frac{2}{1+\sqrt{5}} \cdot \frac{\sqrt{5}-1}{\sqrt{5}-1} = \frac{2(\sqrt{5}-1)}{4} = \frac{\sqrt{5}-1}{2},$$

which agrees. Therefore

$$\cos\frac{2\pi}{5} = \frac{1}{2d} = \frac{1}{2}(d-1) = \frac{1}{2}\cdot\frac{\sqrt{5}-1}{2} = \boxed{\frac{\sqrt{5}-1}{4}}.$$

Exercise 6. An isosceles triangle has vertex angle $36^\circ = \pi/5$, base angles $72^\circ = 2\pi/5$, and base length 4. Find its equal side lengths and its altitude exactly. Explain why changing the scale does not change the cosine of a base angle.

Solution. Let the equal sides have length L. The altitude to the base meets it at its midpoint, forming a right triangle with hypotenuse L and base angle 72° adjacent to the leg $4/2 = 2$. Hence

$$\cos 72^\circ = \frac{2}{L} \quad\Longrightarrow\quad L = \frac{2}{(\sqrt{5}-1)/4} = \frac{8}{\sqrt{5}-1} = \frac{8(\sqrt{5}+1)}{4} = \boxed{2(1+\sqrt{5})}.$$

(This is $4d$, the lecture triangle scaled by 4.) The altitude is

$$h = \sqrt{L^2 - 2^2} = \sqrt{4(6+2\sqrt{5})-4} = \sqrt{20+8\sqrt{5}} = \boxed{2\sqrt{5+2\sqrt{5}}} \approx 6.155.$$

Check: $h = L\sin 72^\circ = 2(1+\sqrt{5})\cdot\dfrac{\sqrt{10+2\sqrt{5}}}{4}$, and squaring gives $\dfrac{(6+2\sqrt{5})(10+2\sqrt{5})}{4} = \dfrac{80+32\sqrt{5}}{4} = 20+8\sqrt{5}$.

Scaling by a factor $s > 0$ multiplies every side by s, so the similar triangles have equal side ratios. Since $\cos 72^\circ = \dfrac{\text{adjacent}}{\text{hypotenuse}}$, the factor s cancels and the cosine is unchanged.

Exercise 7. Use the complementary-angle relations to find exact values of $\sin 18^\circ = \sin(\pi/10)$ and $\cos 18^\circ = \cos(\pi/10)$ from the values at $72^\circ = 2\pi/5$.

Solution. Since $18^\circ = 90^\circ - 72^\circ$, i.e. $\pi/10 = \pi/2 - 2\pi/5$,

$$\sin\frac{\pi}{10} = \sin\left(\frac{\pi}{2}-\frac{2\pi}{5}\right) = \cos\frac{2\pi}{5} = \boxed{\frac{\sqrt{5}-1}{4}} \approx 0.3090,$$

$$\cos\frac{\pi}{10} = \cos\left(\frac{\pi}{2}-\frac{2\pi}{5}\right) = \sin\frac{2\pi}{5} = \boxed{\frac{\sqrt{10+2\sqrt{5}}}{4}} \approx 0.9511.$$

Check: $\left(\frac{\sqrt{5}-1}{4}\right)^2 + \frac{10+2\sqrt{5}}{16} = \frac{6-2\sqrt{5}+10+2\sqrt{5}}{16} = 1$.

Exercise 8. In the triangle ACD from this section, use the sine law to show that

$$\sin\frac{\pi}{5} = \frac{\sin(2\pi/5)}{d}.$$

Simplify this to an exact value for $\sin 36^\circ = \sin(\pi/5)$, then use the Pythagorean identity to find $\cos 36^\circ = \cos(\pi/5)$.

Solution. In triangle ACD, side $CD = 1$ is opposite $\angle A = \pi/5$, and side $AC = d$ is opposite $\angle D = 2\pi/5$. The sine law gives

$$\frac{CD}{\sin(\pi/5)} = \frac{AC}{\sin(2\pi/5)} \quad\Longrightarrow\quad \frac{1}{\sin(\pi/5)} = \frac{d}{\sin(2\pi/5)},$$

so $\sin(\pi/5) = \sin(2\pi/5)/d$. Using $1/d = (\sqrt{5}-1)/2$,

$$\sin\frac{\pi}{5} = \frac{\sqrt{10+2\sqrt{5}}}{4}\cdot\frac{\sqrt{5}-1}{2} = \frac{(\sqrt{5}-1)\sqrt{10+2\sqrt{5}}}{8}.$$

To put this in simpler form, square it:

$$\frac{(6-2\sqrt{5})(10+2\sqrt{5})}{64} = \frac{60+12\sqrt{5}-20\sqrt{5}-20}{64} = \frac{40-8\sqrt{5}}{64} = \frac{5-\sqrt{5}}{8} = \frac{10-2\sqrt{5}}{16}.$$

Since $\sin 36^\circ > 0$,

$$\boxed{\sin 36^\circ = \sin\frac{\pi}{5} = \frac{\sqrt{10-2\sqrt{5}}}{4}} \approx 0.5878.$$

Then

$$\cos^2\frac{\pi}{5} = 1 - \frac{10-2\sqrt{5}}{16} = \frac{6+2\sqrt{5}}{16} = \left(\frac{1+\sqrt{5}}{4}\right)^2,$$

and the cosine of an acute angle is positive, so

$$\boxed{\cos 36^\circ = \cos\frac{\pi}{5} = \frac{1+\sqrt{5}}{4}} = \frac{d}{2} \approx 0.8090.$$

Section 3: Sine and Cosine Graphs

Exercise 9. Sketch $y = \sin x$ and $y = \cos x$ on $[-2\pi, 2\pi]$. Mark the zeros, maxima, and minima.

Solution. From the table of key points and periodicity:

- $\sin x$: zeros $-2\pi, -\pi, 0, \pi, 2\pi$; maxima (value 1) at $-3\pi/2$ and $\pi/2$; minima (value -1) at $-\pi/2$ and $3\pi/2$.
- $\cos x$: zeros $-3\pi/2, -\pi/2, \pi/2, 3\pi/2$; maxima (value 1) at $-2\pi, 0, 2\pi$; minima (value -1) at $-\pi$ and π.

Exercise 10. List the intervals on which $\sin x$ is positive and those on which it is negative for $0 \le x \le 2\pi$. Do the same for $\cos x$.

Solution. The sine is the y-coordinate on the unit circle, so

$$\sin x > 0 \text{ on } \boxed{(0,\pi)}, \qquad \sin x < 0 \text{ on } \boxed{(\pi, 2\pi)}.$$

The cosine is the x-coordinate, so

$$\cos x > 0 \text{ on } \boxed{[0, \tfrac{\pi}{2}) \cup (\tfrac{3\pi}{2}, 2\pi]}, \qquad \cos x < 0 \text{ on } \boxed{(\tfrac{\pi}{2}, \tfrac{3\pi}{2})}.$$

The zeros ($x = 0, \pi, 2\pi$ for sine and $x = \pi/2, 3\pi/2$ for cosine) are in neither list.

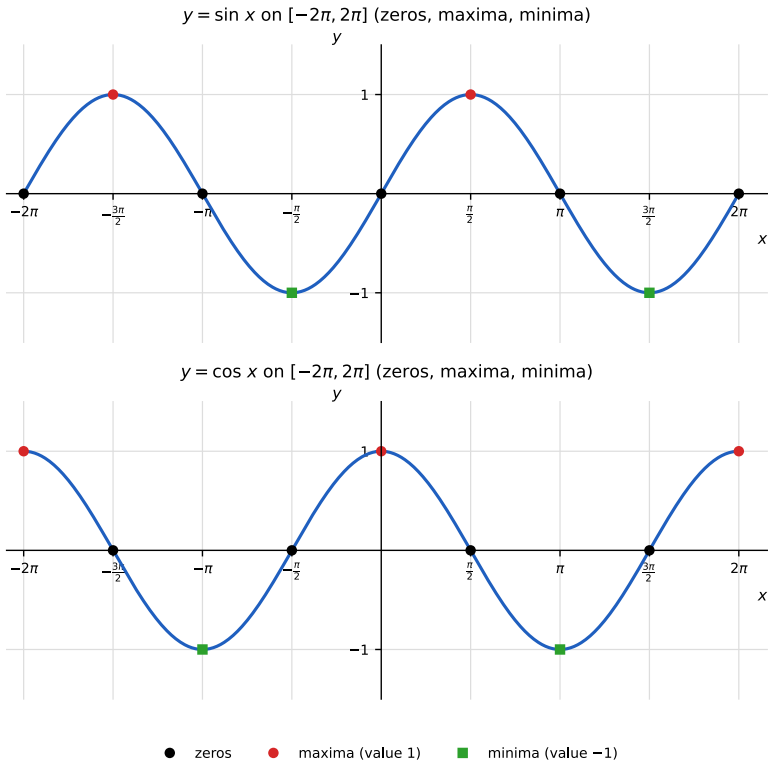


Figure 1: Graphs of y equals sin x and y equals cos x on minus 2 pi to 2 pi with zeros, maxima, and minima marked

Exercise 11. Explain why $y = \sin x$ and $y = \sin(x + 2\pi)$ have the same graph. How does $y = \sin(x + \pi)$ differ from $y = \sin x$?

Solution. Adding 2π to an angle is one full turn of the unit circle, returning to the same point, so $\sin(x + 2\pi) = \sin x$ for every x. Two functions with equal values at every input have the same graph; the left shift by 2π is a whole period and moves the graph onto itself.

Adding π is a half turn, which sends $(\cos x, \sin x)$ to $(-\cos x, -\sin x)$. Thus $\sin(x+\pi) = -\sin x$: its graph is the sine graph shifted left by π, which is the same as its reflection across the x-axis. For example, at $x = \pi/2$ the first function equals 1 while $\sin(3\pi/2) = -1$.

Exercise 12. Decide whether each function is even, odd, or neither: $\cos x + 2$, $\sin x + 2$, $\sin x \cos x$, and $\sin^2 x$.

Solution. Use $\cos(-x) = \cos x$ and $\sin(-x) = -\sin x$.

- $f(x) = \cos x + 2$: $f(-x) = \cos x + 2 = f(x)$, so it is **even**.
- $g(x) = \sin x + 2$: $g(-x) = -\sin x + 2$. This is not $g(x)$ in general and not $-g(x) = -\sin x - 2$ either. For example, $g(\pi/2) = 3$, $g(-\pi/2) = 1$, which is neither 3 nor -3. So it is **neither**.
- $h(x) = \sin x \cos x$: $h(-x) = (-\sin x)(\cos x) = -h(x)$, so it is **odd**.
- $p(x) = \sin^2 x$: $p(-x) = (-\sin x)^2 = \sin^2 x = p(x)$, so it is **even**.

Section 4: Tangent and the Reciprocal Functions

Exercise 13. Sketch two periods of $y = \tan x$. State its domain, range, zeros, and vertical asymptotes.

Solution. The period is π, so two periods fit on $(-\pi/2, 3\pi/2)$. Each branch is increasing, passes through $(k\pi, 0)$, and through $(k\pi \pm \pi/4, \pm 1)$.

- Domain: $\boxed{x \neq \pi/2 + k\pi}$.
- Range: $\boxed{\mathbb{R}}$.
- Zeros: $\boxed{x = k\pi}$.
- Vertical asymptotes: $\boxed{x = \pi/2 + k\pi}$ (in the sketch, $x = -\pi/2, \pi/2, 3\pi/2$).

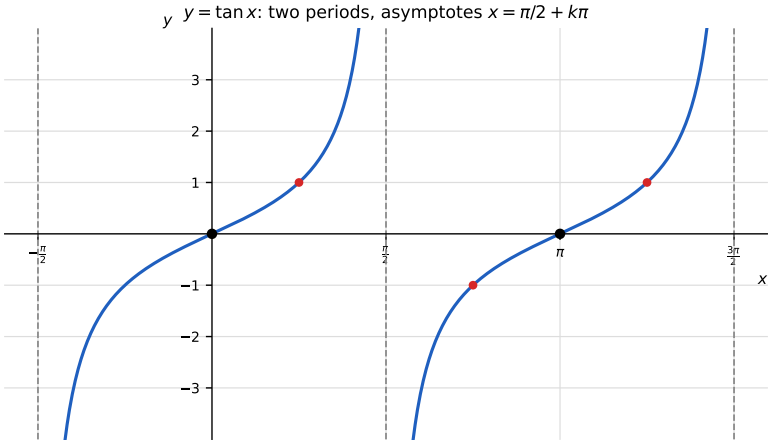


Figure 2: Graph of y equals tan x on minus pi over 2 to 3 pi over 2 with vertical asymptotes at odd multiples of pi over 2

Exercise 14. Sketch $y = \sec x$ and $y = \csc x$ separately on $[0, 2\pi]$, using cosine and sine as guides. Mark the asymptotes.

Solution. For $\sec x = 1/\cos x$, the asymptotes are at the zeros of cosine, $x = \pi/2$ and $x = 3\pi/2$. The graph passes through $(0, 1)$, $(\pi, -1)$, and $(2\pi, 1)$, with branches above 1 on $[0, \pi/2)$ and $(3\pi/2, 2\pi]$ and a branch below -1 on $(\pi/2, 3\pi/2)$.

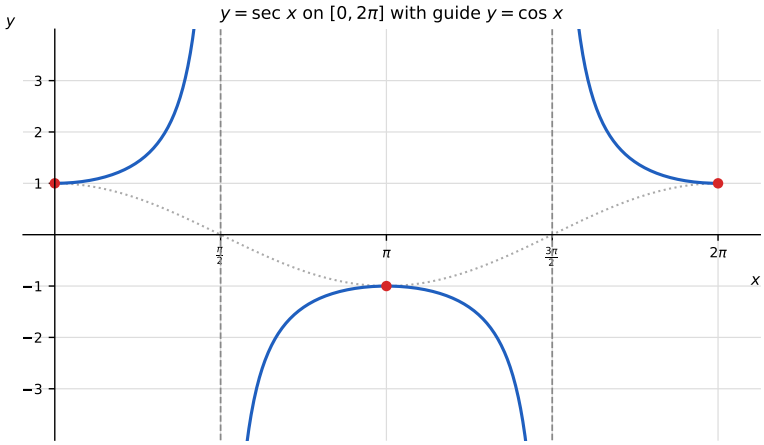


Figure 3: Graph of y equals sec x on 0 to 2 pi with asymptotes at pi over 2 and 3 pi over 2

For $\csc x = 1/\sin x$, the asymptotes are at the zeros of sine, $x = 0, \pi, 2\pi$. The graph passes through $(\pi/2, 1)$ and $(3\pi/2, -1)$, with one branch above 1 on $(0, \pi)$ and one below -1 on $(\pi, 2\pi)$.

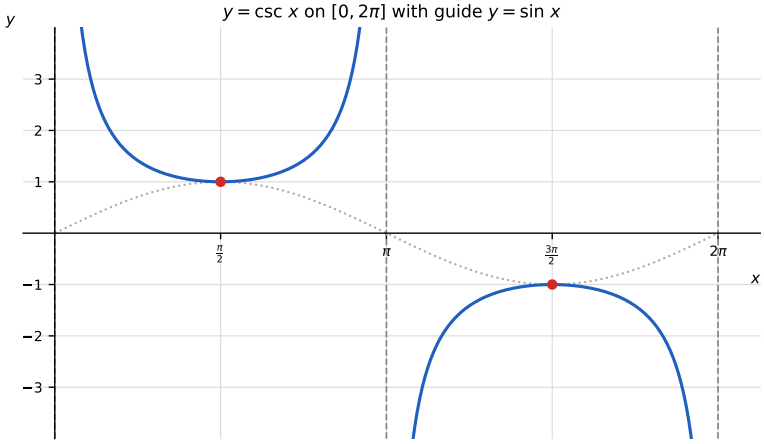


Figure 4: Graph of y equals csc x on 0 to 2 pi with asymptotes at 0, pi, and 2 pi

Exercise 15. Explain why $\cot x = 1/\tan x$ holds only where both sides are defined. Give an input at which $\cot x$ is defined but $1/\tan x$ is not.

Solution. The identity $\cot x = \cos x/\sin x = 1/(\sin x/\cos x)$ requires $\cos x \neq 0$ so that $\tan x$ exists, and $\tan x \neq 0$ so that we may divide by it. The right side $1/\tan x$ is therefore defined exactly when $x \neq \pi/2 + k\pi$ and $x \neq k\pi$, while $\cot x$ is defined whenever $\sin x \neq 0$, i.e. $x \neq k\pi$. They differ at $x = \pi/2 + k\pi$.

For example, at $x = \pi/2$: $\cot(\pi/2) = \cos(\pi/2)/\sin(\pi/2) = 0/1 = \boxed{0}$ is defined, but $\tan(\pi/2)$ is undefined, so $1/\tan(\pi/2)$ is not defined.

Section 5: Transformations

Exercise 16. For $y = -2\cos(3x) + 1$, state the amplitude, period, midline, maximum, and minimum, and sketch one full period.

Solution. Here $A = -2$, $B = 3$, $C = 0$, $D = 1$.

- Amplitude: $\boxed{2}$.
- Period: $T = 2\pi/3$, so $\boxed{2\pi/3}$.
- Midline: $\boxed{y = 1}$.
- Maximum: $1 + 2 = \boxed{3}$; minimum: $1 - 2 = \boxed{-1}$.

Because $A < 0$, the graph is an upside-down cosine, starting at its minimum. With $T/4 = \pi/6$, the five key points are

$$(0, -1), \quad \left(\tfrac{\pi}{6}, 1\right), \quad \left(\tfrac{\pi}{3}, 3\right), \quad \left(\tfrac{\pi}{2}, 1\right), \quad \left(\tfrac{2\pi}{3}, -1\right).$$

Check: $-2\cos(3 \cdot \pi/3) + 1 = -2(-1) + 1 = 3$, and $-2\cos(\pi/2) + 1 = 1$ at $x = \pi/6$. The maximum occurs at $x = \pi/3 + 2k\pi/3$.

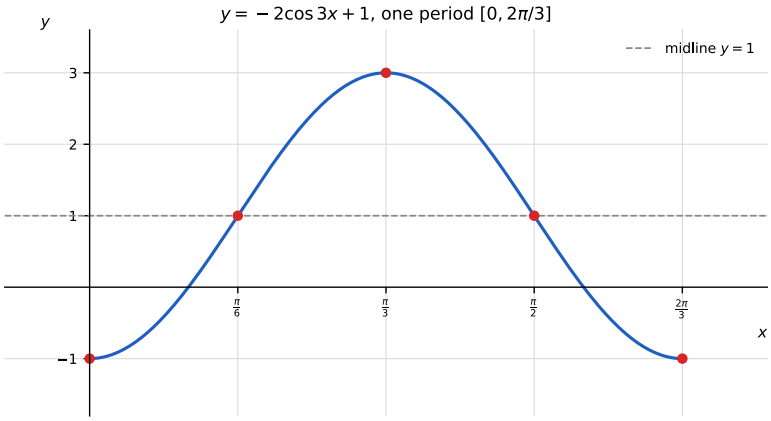


Figure 5: Graph of y equals minus 2 cos 3x plus 1 over one period from 0 to 2 pi over 3 with the midline y equals 1

Exercise 17. For $y = 2\sin(x - \pi/3) - 1$, state the amplitude, period, phase shift, and midline, and sketch one period marking five key points.

Solution. Here $A = 2$, $B = 1$, $C = \pi/3$, $D = -1$.

- Amplitude: $\boxed{2}$; period: $\boxed{2\pi}$.
- Phase shift: $\boxed{\pi/3 \text{ to the right}}$; midline: $\boxed{y = -1}$.

Start at $x = C = \pi/3$ and step by $T/4 = \pi/2$. For sine, the values are $D, D + A, D, D - A, D$:

$$\left(\tfrac{\pi}{3}, -1\right), \quad \left(\tfrac{5\pi}{6}, 1\right), \quad \left(\tfrac{4\pi}{3}, -1\right), \quad \left(\tfrac{11\pi}{6}, -3\right), \quad \left(\tfrac{7\pi}{3}, -1\right).$$

Check, e.g., $2\sin(5\pi/6 - \pi/3) - 1 = 2\sin(\pi/2) - 1 = 1$ and $2\sin(11\pi/6 - \pi/3) - 1 = 2\sin(3\pi/2) - 1 = -3$.

Exercise 18. Graph $y = \cos(2x + \pi/2)$ for $-\pi \le x \le \pi$. Identify the period and phase shift.

Solution. Factor the inside: $2x + \pi/2 = 2(x + \pi/4)$. So $B = 2$, $C = -\pi/4$:

$$\text{period} = \frac{2\pi}{2} = \boxed{\pi}, \qquad \text{phase shift} = \boxed{\pi/4 \text{ to the left}}.$$

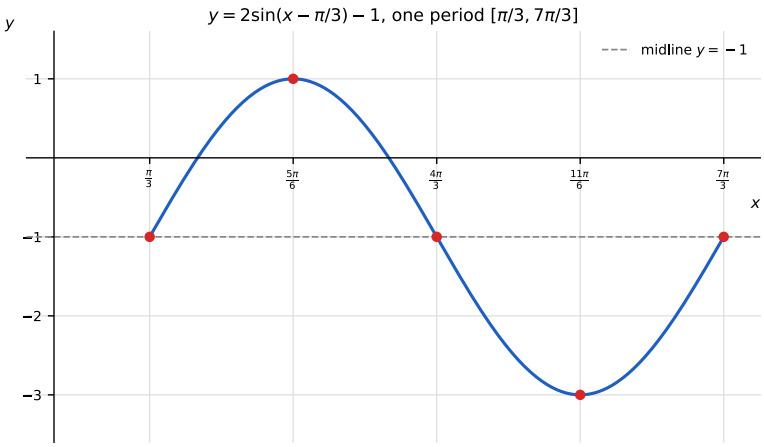


Figure 6: Graph of y equals 2 sin of x minus pi over 3 minus 1 over one period from pi over 3 to 7 pi over 3 with five key points and the midline y equals minus 1

The cycle starts at $x = -\pi/4$ with the maximum 1, and with $T/4 = \pi/4$ the key points are $(-\pi/4, 1)$, $(0, 0)$, $(\pi/4, -1)$, $(\pi/2, 0)$, $(3\pi/4, 1)$; the pattern repeats every π. On $[-\pi, \pi]$ there are two full periods. (Equivalently, $\cos(2x + \pi/2) = -\sin 2x$, since $\cos(u + \pi/2) = -\sin u$.)

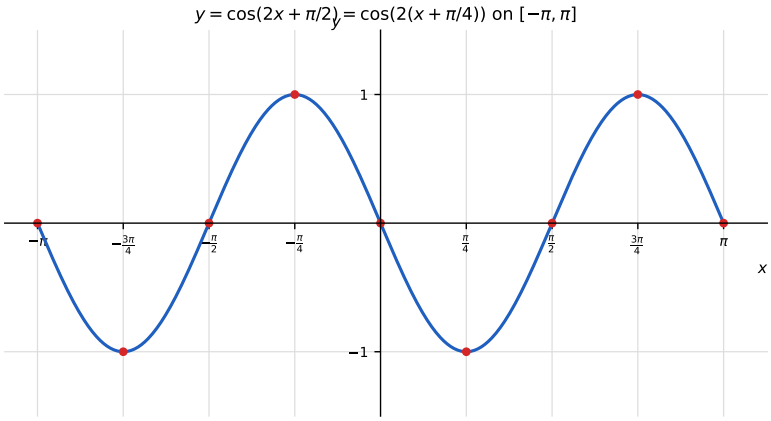


Figure 7: Graph of y equals cos of 2x plus pi over 2 on minus pi to pi, with maxima at minus pi over 4 and 3 pi over 4 and minima at minus 3 pi over 4 and pi over 4

Exercise 19. State the period and vertical asymptotes of $y = \tan(2x)$, and sketch two periods. Explain why its amplitude is undefined.

Solution. The period is $\pi/|B| = \boxed{\pi/2}$. Asymptotes occur where $2x = \pi/2 + k\pi$, i.e.

$$\boxed{x = \frac{\pi}{4} + \frac{k\pi}{2}}.$$

Two periods fit on $(-\pi/4, 3\pi/4)$. The branches pass through $(-\pi/8, -1)$, $(0, 0)$, $(\pi/8, 1)$ and $(3\pi/8, -1)$, $(\pi/2, 0)$, $(5\pi/8, 1)$.

Amplitude is the maximum distance from the midline, but tangent has range $\mathbb{R}$ and no maximum or minimum value, so no such distance exists.

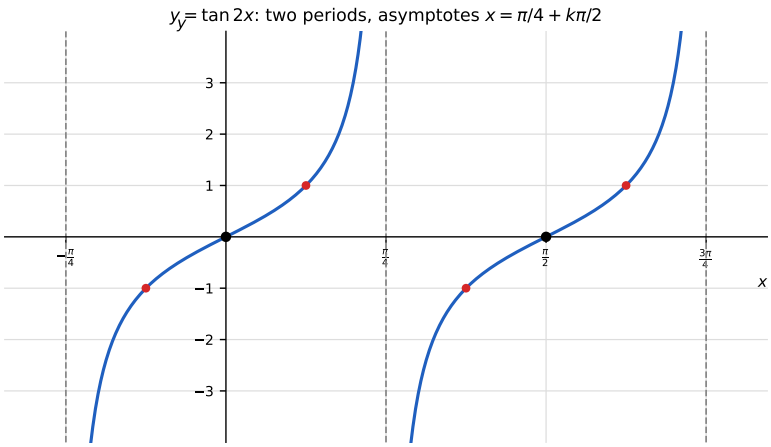


Figure 8: Graph of y equals tan 2x over two periods with vertical asymptotes at minus pi over 4, pi over 4, and 3 pi over 4

Exercise 20. Rewrite $y = -3\sin(-2x + \pi) + 2$ with a positive coefficient of x inside the sine. State its amplitude, period, phase shift, and range.

Solution. Since sin is odd, $\sin(-u) = -\sin u$ with $u = 2x - \pi$:

$$y = -3\sin(-(2x - \pi)) + 2 = 3\sin(2x - \pi) + 2 = \boxed{3\sin\left(2\left(x - \tfrac{\pi}{2}\right)\right) + 2}.$$

Thus the amplitude is $\boxed{3}$, the period is $2\pi/2 = \boxed{\pi}$, the phase shift is $\boxed{\pi/2 \text{ to the right}}$, and the midline is $y = 2$, so the range is $\boxed{[-1, 5]}$. Check at $x = \pi/4$: the original gives $-3\sin(-\pi/2+\pi)+2 = -3\sin(\pi/2)+2 = -1$, and the new form gives $3\sin(-\pi/2) + 2 = -1$.

Exercise 21. Find all x where $y = 2\cos(3x - \pi) + 4$ reaches its maximum, and all x where it reaches its minimum.

Solution. The range is $[4 - 2, 4 + 2] = [2, 6]$. The maximum 6 occurs when $\cos(3x - \pi) = 1$:

$$3x - \pi = 2k\pi \implies x = \frac{\pi}{3} + \frac{2k\pi}{3} = \boxed{\frac{(2k+1)\pi}{3}}.$$

The minimum 2 occurs when $\cos(3x - \pi) = -1$:

$$3x - \pi = \pi + 2k\pi \implies x = \frac{2\pi}{3} + \frac{2k\pi}{3} = \boxed{\frac{2(k+1)\pi}{3}} \quad (\text{that is, } x = \tfrac{2m\pi}{3},\ m \in \mathbb{Z}).$$

Check: $x = \pi/3$ gives $2\cos(0) + 4 = 6$; $x = 0$ gives $2\cos(-\pi) + 4 = 2$.

Section 6: Identities and Equations

Exercise 22. Solve on $[0, 2\pi)$: $2\cos x + 1 = 0$, $\tan x = \sqrt{3}$, and $\sin x = \cos x$.

Solution.

- $2\cos x + 1 = 0$ gives $\cos x = -1/2$. The reference angle is $\pi/3$, and cosine is negative in quadrants II and III:

$$\boxed{x = \frac{2\pi}{3},\ \frac{4\pi}{3}}.$$

- $\tan x = \sqrt{3}$. The reference angle is $\pi/3$, and tangent is positive in quadrants I and III:

$$\boxed{x = \frac{\pi}{3},\ \frac{4\pi}{3}}.$$

- $\sin x = \cos x$. Here $\cos x \neq 0$ (if $\cos x = 0$ then $\sin x = \pm 1 \neq 0$), so we may divide: $\tan x = 1$, giving

$$\boxed{x = \frac{\pi}{4},\ \frac{5\pi}{4}}.$$

Checks: $\cos(2\pi/3) = -1/2$; $\tan(4\pi/3) = \sin(4\pi/3)/\cos(4\pi/3) = \sqrt{3}$; $\sin(5\pi/4) = \cos(5\pi/4) = -\sqrt{2}/2$.

Exercise 23. Find all real solutions of $\cos x = 0$ and $\tan x = -1$.

Solution. $\cos x = 0$ at the left and right ends of the unit circle:

$$\boxed{x = \frac{\pi}{2} + k\pi}.$$

For $\tan x = -1$, the reference angle is $\pi/4$ and the solutions in $[0, \pi)$ are $3\pi/4$ only (tangent is negative in quadrant II). Since tan has period π,

$$\boxed{x = \frac{3\pi}{4} + k\pi} \quad \left(\text{equivalently } x = -\frac{\pi}{4} + k\pi\right).$$

These satisfy $\cos x \neq 0$, so $\tan x$ is defined there.

Exercise 24. Solve $2\sin(2x - \pi/3) = \sqrt{3}$ on $[0, 2\pi)$. State the interval for the inner angle before solving.

Solution. Let $u = 2x - \pi/3$. Since $0 \leq x < 2\pi$, we have $0 \leq 2x < 4\pi$ and

$$-\frac{\pi}{3} \leq u < \frac{11\pi}{3},$$

an interval of length 4π (two cycles). The equation is $\sin u = \sqrt{3}/2$, with solutions $u = \pi/3 + 2n\pi$ and $u = 2\pi/3 + 2n\pi$. Those in the interval are

$$u = \frac{\pi}{3},\ \frac{2\pi}{3},\ \frac{7\pi}{3},\ \frac{8\pi}{3}$$

(the next ones, $13\pi/3$ and $14\pi/3$, are too large; the previous ones, $-5\pi/3$ and $-4\pi/3$, are too small). Solving $x = (u + \pi/3)/2$:

$$\boxed{x = \frac{\pi}{3},\ \frac{\pi}{2},\ \frac{4\pi}{3},\ \frac{3\pi}{2}}.$$

Check: $x = \pi/3$ gives $2\sin(\pi/3) = \sqrt{3}$; $x = 3\pi/2$ gives $2\sin(3\pi - \pi/3) = 2\sin(8\pi/3) = 2\sin(2\pi/3) = \sqrt{3}$.

Exercise 25. Solve $2\cos^2 x + \cos x - 1 = 0$ on $[0, 2\pi)$.

Solution. Factor as a quadratic in $\cos x$:

$$(2\cos x - 1)(\cos x + 1) = 0.$$

Then $\cos x = 1/2$ gives $x = \pi/3,\ 5\pi/3$, and $\cos x = -1$ gives $x = \pi$:

$$\boxed{x = \frac{\pi}{3},\ \pi,\ \frac{5\pi}{3}}.$$

Check at $x = \pi$: $2(1) + (-1) - 1 = 0$.

Exercise 26. Simplify $\dfrac{1-\sin^2 x}{\cos x}$ and $\dfrac{\sec^2 x - 1}{\tan x}$, and state the domain of each original expression.

Solution. Since $1 - \sin^2 x = \cos^2 x$,

$$\frac{1-\sin^2 x}{\cos x} = \frac{\cos^2 x}{\cos x} = \boxed{\cos x}, \qquad \text{domain: } x \neq \frac{\pi}{2} + k\pi,$$

because the original is undefined where $\cos x = 0$.

Since $\sec^2 x - 1 = \tan^2 x$,

$$\frac{\sec^2 x - 1}{\tan x} = \frac{\tan^2 x}{\tan x} = \boxed{\tan x}.$$

The original needs $\sec x$ and $\tan x$ defined, so $x \neq \pi/2 + k\pi$, and it needs $\tan x \neq 0$, so $x \neq k\pi$. Combined,

$$\text{domain: } \boxed{x \neq \frac{k\pi}{2} \ (k \in \mathbb{Z})}.$$

The simplified formula $\tan x$ is defined (and equals 0) at $x = k\pi$, but those points are not in the original domain.

Exercise 27. Solve $\sin x \cos x = \sin x$ on $[0, 2\pi)$. Explain which solutions would be lost by dividing by $\sin x$.

Solution. Move everything to one side and factor:

$$\sin x \cos x - \sin x = \sin x(\cos x - 1) = 0.$$

So $\sin x = 0$, giving $x = 0, \pi$, or $\cos x = 1$, giving $x = 0$. The solutions are

$$\boxed{x = 0,\ \pi}.$$

Dividing by $\sin x$ is allowed only where $\sin x \neq 0$. It leaves $\cos x = 1$ and discards the case $\sin x = 0$ entirely, which contains the solutions $x = 0$ and $x = \pi$. The solution $x = \pi$ would be lost completely, since $\cos \pi = -1 \neq 1$ (and $x = 0$ would be found only because it also solves $\cos x = 1$). Check: at $x = \pi$, both sides equal 0.